Requirements Document: AI-Enabled Skills and Contribution Assessor (POC Phase)

Project Overview

This project aims to create an AI-enabled system leveraging the GitHub Issue API and ChatGPT to summarize intern contributions and classify skills gained per issue in the Hack for LA organization. It generates resume-ready bullet points and adds skill set labels from a predefined CSV taxonomy.

Functional Requirements (POC Scope)

User Input and Interaction

- A predefined list of contributors will be used (no text entry or user input in the POC).

Data Processing

- Fetch comments per issue for each user in the predefined list via GitHub Issue API.
- Extract context from issue overview and action items.
- Submit combined issue context and user comments to ChatGPT.

Output Generation

- **Bullet Points:**

- Formal style with software development concepts.

- Use the STAR (Situation, Task, Action, Result) method for clarity.
- Maximum of 30 words per bullet point.
- Explicitly state task status (for POC, all issues are closed).

- Resume bullet comment will follow a standardized format:

    - Opening message describing the purpose, with a link to more detail about the automation.
    - Bullet points grouped per user, e.g.:

    ```
    @username1
    - Resume Bullet 1
    - Resume Bullet 2

    @username2
    - Resume Bullet 1
    - Resume Bullet 2
    ```

- **Skill Classification:**

- Based on provided CSV file (`HfLA_skills.csv`), structured as comma-separated labels.

- Multiple skill labels will be assigned per user.
- Label comment formatting:
    - Opening message
    - Markdown checklist of suggested skills:

    ```
    - [ ] Python
    - [ ] Project Management
    ```

Non-Functional Requirements

Performance

- The entire operation per issue per user must complete within 1 minute.

Reliability and Error Handling

- Errors from GitHub or OpenAI APIs:
- Publicly post the error code and user-friendly message to the original issue.
- Log error details to a structured error log file.
- These error logs will help identify persistent or pattern-based failures.

Security

- No encryption or anonymization required.
- No anticipated privacy concerns.

Deployment and Environment

- No existing deployment infrastructure.
- Runs daily across all open issues in the Hack for LA organization using a time-triggered script.
- GitHub is used as the database; output files will be written in **CSV and JSON** formats under the `/Data` folder.
- Schema for these files is TBD but will support:
- Contributor ID and name
- Associated issue ID
- Resume bullets
- Skill labels

Scalability

- Designed for low concurrent load via sequential batch processing.

Usability

- No GUI; automated backend process using scripts.

Technical Stack and System Notes

- The original JavaScript codebase (`true-github-contributors`) is being rewritten in **Python**, and all development will use Python going forward.

Data and API Requirements

GitHub API

- Uses GitHub Issue API to fetch comments and post results.
- Requires a bot account with proper collaborator permissions (avoids spam restrictions).
- Uses a **Personal Access Token** (configured to never expire) for authentication.
- Related documentation:
- Hack for LA Website Wiki
- HfLA-Website-Admin: How to update Hackforlabot token

OpenAI API

- GPT-4 Turbo or GPT-4o will be used.
- OpenAI rate limits (Tier 1):
- **RPM (Requests per Minute):** 500
- **TPM (Tokens per Minute):** 30,000
- These limits will be respected using throttling logic.
- Excessive token use will be avoided through input size control.

Audit and Maintenance

- Log all bot activity, including:
- User mentions
- Generated content
- Errors and diagnostics
- Rotate access tokens periodically.
- Monitor GitHub and OpenAI dashboards for outages.

Future Considerations (Post-POC / Pilot Phase)

- Accepting manual contributor input via UI or CLI
- Handling ongoing (open) issues
- More granular user notification (e.g., DM vs. group mention)
- Schema definition and persistence via database or versioned storage
- Analytics for skill acquisition trends across org

This revised document defines the POC phase of the AI Skills Assessor project. It captures constraints, goals, and structural considerations needed to implement, evaluate, and shape a full-scale pilot.